

NTRO Signal Analyzer — Complete Technical Guide

A Comprehensive Engineering Reference for SIH 2026 Problem Statement 26147 *Autonomous RF Signal Parameter Extraction for the National Technical Research Organisation*

Authors: Signal Intelligence Engineering Team (ECE, 5th Semester) **Target:** Smart India Hackathon 2026 Evaluators & Engineering Students **Edition:** 3.0 (Architecture v2 — Adaptive Pipeline, Epistemic Hierarchy, Open-Set Rejection)

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1. The Problem We Are Solving

1.1 The SIH 2026 Challenge (SIH26147)

Smart India Hackathon 2026 Problem Statement **SIH26147**, submitted by the **National Technical Research Organisation (NTRO)**, requires:

"Development of a software solution for autonomous parameter extraction of intercepted RF signals without prior knowledge of signal parameters."

In plain language: a monitoring station or surveillance aircraft intercepts an unknown radio signal. The operator has no manual labels, no transmission logs, and no prior knowledge of who is transmitting or what hardware they are using. The software must ingest the raw recording, classify the modulation format, extract all physical transmission parameters, and deliver structured intelligence ready for tactical decoders.

+-----+		+-----+		+-----	
---+					
Intercepted RF	--->	Autonomous Parameter	--->	Tactical	
Intelligence					
Raw Recording		Extraction Engine		- Carrier: 2.13 kHz	
(.wav / .iq)		(NTRO Signal Analyzer)		- 99% OBW: 323 Hz	
+-----+		+-----+		- Mode: 2-FSK	
NAVTEX					
				- Baud: 100.0 Baud	
				- SNR: 11.7 dB	
				+-----	
---+					

1.2 The Real-World SIGINT Problem

In a college laboratory, students work with known signals: a function generator outputs a clean 1 kHz sine wave, or a simulation script generates BPSK at a preset carrier.

In real-world signals intelligence (SIGINT) and electronic warfare (EW), you face an open radio spectrum. The antenna intercepts:

- High-frequency maritime telex transmissions (NAVTEX, SITOR-B)
- Over-the-horizon radar pulses searching for aircraft over thousands of kilometers
- Military tactical link transmissions (MIL-STD-188-141 ALE)
- Amateur radio emergency messages (FT8, PSK31)
- Cellular base station broadcasts (GSM, DMR)
- Satellite weather faxes (WEFAX)
- Satellite telemetry beacons from orbiting CubeSats

Each signal occupies a different slice of bandwidth, transmits at a different speed, and modulates electromagnetic waves using different physical principles.

1.3 Why Traditional Approaches Fail

Previous attempts at automated signal parameter extraction fall into two traps:

1. **Manual Selection Dropdowns:** Older tools require the user to pick "FSK" or "Radar" from a dropdown menu and manually type the sampling rate. This is useless for autonomous intercept systems where no human operator is present.
2. **Brittle Neural Networks:** Deep learning classifiers trained on synthetic datasets fail when exposed to real-world multipath fading, soundcard anti-aliasing filter roll-offs, and varying receiver audio beat frequencies. A neural network cannot explain why it picked a class, and it frequently produces false positives with high confidence.

1.4 What Our System Accomplishes

Our engine relies on **deterministic physical invariants and multi-stage evidence fusion**. Instead of guessing through neural weights or a lookup table, it builds a hierarchy of evidence:

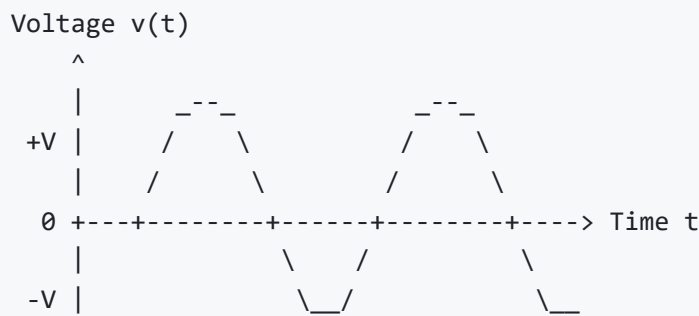
1. Extracts multi-domain physical features (spectral, envelope, instantaneous frequency, cumulants).
2. Runs a Gaussian noise pre-gate to prevent any named protocol from being returned when input is noise.
3. Tests physical invariant rules in strict priority order.
4. Tags every parameter with an **epistemic status** (OBSERVED, ESTIMATED, HYPOTHESIZED, VALIDATED, UNKNOWN).
5. Returns `UNKNOWN` when evidence is insufficient rather than forcing a named class.
6. Runs a full physical reconstruction chain (IQ conditioning → 20-D features → open-set AMC → sync → demod → FEC → framing → evidence fusion) to build secondary evidence.

This pipeline is a **hybrid DSP/physics-based architecture with open-set rejection and evidence fusion**, not a simple lookup table.

2. Signals in the Real World — DSP Fundamentals

2.1 What Actually is a Signal?

An electrical signal is a voltage that changes over time. When an electromagnetic wave strikes an antenna, it induces a tiny oscillating voltage in the metal rod:



An Analog-to-Digital Converter (ADC) samples this continuous voltage at regular intervals:

Continuous: $v(t)$
 Sampled: $x[n] = v(n \cdot T_s) = v(n / f_s)$

Where:

- f_s = Sampling frequency (samples per second, in Hz)
- $T_s = 1 / f_s$ = Sampling period (seconds between consecutive samples)
- n = Integer sample index (0, 1, 2, ...)

2.2 Real Signals vs. Complex Baseband (I/Q Representation)

A real-valued signal consists of a single voltage value at each time step:

$$x(t) = A(t) \cdot \cos(2\pi \cdot f_c \cdot t + \phi(t))$$

Real-life Analogy: Imagine tracking a car moving on a straight line back and forth. You only need one number (distance from origin) to describe its position.

However, high-frequency radio carriers oscillate millions of times per second. To capture both the amplitude and the phase direction without tracking billions of oscillations directly, modern Software-Defined Radios (SDRs) split the incoming RF carrier into two separate paths:

1. One path multiplied by $\cos(2\pi \cdot f_c \cdot t)$ → **In-Phase channel (I)**
2. One path multiplied by $-\sin(2\pi \cdot f_c \cdot t)$ → **Quadrature channel (Q)**

```

      +---> Multiplied by  $\cos(2\pi \cdot f_c \cdot t)$  ---> Lowpass Filter ---> In-Phase I(t)
Incoming RF -|
      +---> Multiplied by  $-\sin(2\pi \cdot f_c \cdot t)$  --> Lowpass Filter ---> Quadrature Q(t)
  
```

Together, these form a single **complex baseband number** at each instant:

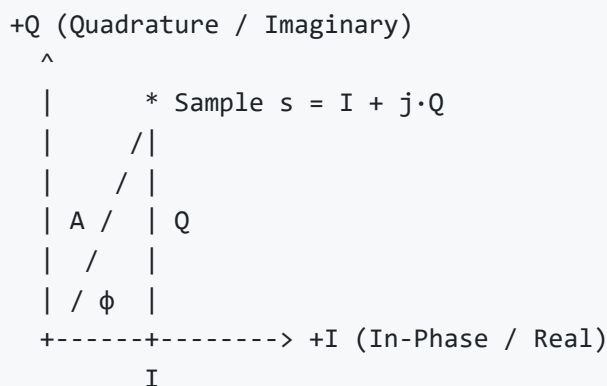
$$s(t) = I(t) + j \cdot Q(t)$$

Where $j = \sqrt{-1}$.

Real-life Analogy: Imagine a clock hand rotating around the center of a clock face.

- I is the horizontal position of the hand tip (x-axis).
- Q is the vertical position of the hand tip (y-axis).
- The length of the hand is the **instantaneous amplitude**: $A = \sqrt{I^2 + Q^2}$.
- The angle of the hand is the **instantaneous phase**: $\phi = \arctan2(Q, I)$.

By having both I and Q , you know exactly where the hand points at every instant, whether it is rotating clockwise or counterclockwise (positive vs. negative frequency), and how fast it rotates.



2.3 The Hilbert Transform & Analytic Signals

Many intercepted recordings come from the audio headphone jack of an HF receiver. These are single-channel, real-valued audio files (`.wav`).

To analyze these with complex I/Q algorithms, we generate an artificial Quadrature channel by shifting every frequency component of the real signal by exactly -90° :

$$s(t) = x(t) + j \cdot H\{ x(t) \}$$

Where $H\{ x(t) \}$ is the **Hilbert Transform**:

$$H\{ x(t) \} = (1 / \pi) \cdot \int [x(\tau) / (t - \tau)] d\tau$$

This constructs the **analytic signal**. In the frequency domain, the negative frequency components cancel out to zero, leaving energy only on the positive frequency axis.

2.4 The Nyquist-Shannon Sampling Theorem & Aliasing

The most famous rule in digital signal processing states:

$$f_s \geq 2 \cdot f_{\max}$$

The sampling rate f_s must be at least twice the highest frequency $f_{\max}$ present in the analog signal.

Real-life Analogy: If you film a wagon wheel rotating at 30 revolutions per second using a 24-frame-per-second camera, the wheel appears to rotate backward or stand still. This optical illusion is **aliasing**.

In audio: if you sample a 15 kHz whistle with a sampling rate of only 20 kHz, the Nyquist limit is 10 kHz. The 15 kHz whistle folds back across the 10 kHz boundary and appears as a fake tone at $20 - 15 = 5$ kHz !

In our system:

- For CD-quality WAV files, $f_s = 44,100$ Hz , giving a Nyquist bandwidth of 22,050 Hz .
- For SDR captures, f_s can reach 2,000,000 Hz (2 MSPS) or 20,000,000 Hz (20 MSPS).

2.5 Modulation Families: How Information Travels Over Air

A pure, unmodulated sine wave carries zero information. To transmit data, the transmitter modulates one or more properties of the carrier:

Modulation Type	What Changes?	Typical Real-World Applications
CW (Continuous Wave)	None (pure carrier tone)	Beacons, frequency references, radar calibration
OOK (On-Off Keying)	Carrier turns ON and OFF	Morse code telegraphy, simple garage remotes
AM (Amplitude Modulation)	Carrier amplitude scales with voice	Aviation communications, commercial AM broadcast
FM (Frequency Modulation)	Carrier frequency shifts smoothly	Walkie-talkies (NFM), weather fax (WEFAX), marine VHF
2-FSK (2-Frequency Shift Keying)	Jumps between 2 discrete frequencies	NAVTEX maritime safety, ASCII teleprinters, RTTY
M-FSK (Multi-Frequency FSK)	Jumps between 4, 8, or 16 discrete tones	FT8 weak-signal amateur radio, 2G ALE military radio
BPSK (Binary Phase Shift Keying)	Carrier phase flips by 180° (0° vs 180°)	PSK31 amateur radio, satellite downlinks
QPSK (Quadrature PSK)	Carrier phase jumps between 4 angles (45°, 135°, 225°, 315°)	Satellite digital video (DVB-S), GPS navigation signals
8-PSK	Carrier phase jumps between 8 angles (45° steps)	STANAG 4285 military naval transmissions
QAM (Quadrature Amplitude Mod)	Both amplitude and phase change (grid points)	Digital cable TV, high-speed WiFi, 4G/5G cellular
Pulsed Radar / FMOP	High-power microsecond bursts with internal chirps	Military early warning radars, coastal sea-state radars

3. The Core Mathematical Toolkit

3.1 The Fourier Transform & FFT Complexity

The Fourier Transform is the mathematical prism of signal processing: it separates a composite signal into its individual sinusoidal frequency components.

The Discrete Fourier Transform (DFT) for N samples is:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j \cdot 2\pi \cdot k \cdot n / N} \quad \text{for } k = 0, 1, \dots, N-1$$

Where:

- $x[n]$ is the input time-domain sample.
- $X[k]$ is the complex spectral coefficient at frequency bin k .
- $e^{-j \cdot 2\pi \cdot k \cdot n / N} = \cos(2\pi \cdot k \cdot n / N) - j \cdot \sin(2\pi \cdot k \cdot n / N)$ via Euler's formula.

Calculating this directly requires N^2 complex multiplications. For $N = 65,536$, that is 4,294,967,296 operations (several seconds of CPU time).

The **Fast Fourier Transform (FFT)** rearranges calculations using symmetries, reducing complexity to:

$$\text{Complexity} = O(N \cdot \log_2(N))$$

For $N = 65,536$, $65,536 \cdot 16 = 1,048,576$ operations — computed in less than 2 milliseconds!

3.2 Power Spectral Density: Welch's Averaged Periodogram

If you take a single raw FFT of a captured radio signal, random noise creates wild sample-to-sample fluctuations. A peak detector on a raw FFT will trigger on random noise spikes.

Peter Welch solved this in 1967 with the **Welch method of averaged modified periodograms**:

```
Time Signal x[n]
[===== Segment 1 =====]
      [===== Segment 2 =====] (50% Overlap)
            [===== Segment 3 =====]
                  ...
```

Each Segment multiplied by Hann Window $w[n]$:

```

      /\
     /\ 
    /\  \
___/    \___ (suppresses edge discontinuity leakage)
```

Compute FFT of each segment ---> Take $|FFT|^2$ ---> Average all segments together

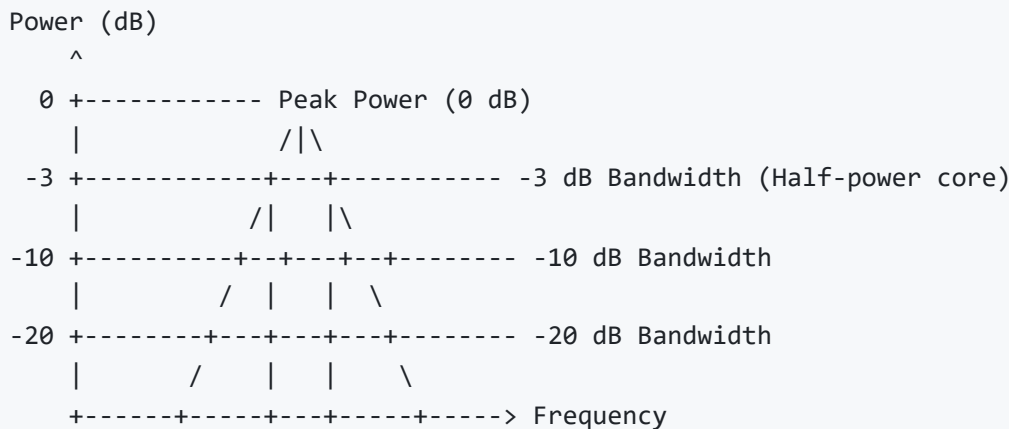
The resulting Power Spectral Density (PSD) is smooth, stationary, and statistically reliable:

$$\text{PSD_dB}[k] = 10 \cdot \log_{10}(\max(P_{\text{linear}}[k], 10^{-18}))$$

Real-life Analogy: A single FFT is like taking one grainy photo at night. Welch's method is like taking 20 rapid photos and averaging them together: the random noise grains average out, while the steady buildings and streetlights stand out razor-sharp.

3.3 Bandwidth Definitions: -3 dB, -10 dB, and 99% Power OBW

How wide is a signal? Radio waves have gradual roll-offs, so bandwidth must be measured against defined physical thresholds:



1. **-3 dB Bandwidth:** The span between frequencies where power drops to half of peak ($10 \cdot \log_{10}(0.5) \approx -3.01 \text{ dB}$).
2. **-10 dB Bandwidth:** The span between frequencies where power drops to 10% of peak ($10 \cdot \log_{10}(0.1) = -10.0 \text{ dB}$).
3. **99% Occupied Power Bandwidth (OBW):** Integrate the linear power from DC to Nyquist:

Cumulative Power: $C[k] = \sum_{i=0}^k P_{\text{linear}}[i]$
 Normalized: $C_{\text{norm}}[k] = C[k] / C[N-1]$

Find bin indices k_{low} where $C_{\text{norm}} = 0.005$ (0.5%) and k_{high} where $C_{\text{norm}} = 0.995$ (99.5%):

$$\text{OBW}_{99\%} = f[k_{\text{high}}] - f[k_{\text{low}}]$$

This captures exactly 99% of total radiated RF energy, regardless of how asymmetric the spectrum is.

3.4 Signal-to-Noise Ratio: Four Physical Regimes

Signal-to-Noise Ratio (SNR) compares desired signal power to background noise:

$$\text{SNR (dB)} = 10 \cdot \log_{10}(P_{\text{signal}} / P_{\text{noise}})$$

A single formula cannot measure SNR across all signal types. Our engine applies three distinct estimators based on signal physics:

Regime 1: M2M4 Continuous Sample Moments (for Continuous Digital PSK/QAM)

Digital communications signals keep a constant or bounded envelope. The received signal is $y[n] = s[n] + w[n]$, where $w[n]$ is Gaussian noise:

$$\begin{aligned} \text{Second Moment: } M_2 &= E[|y[n]|^2] = P_{\text{signal}} + P_{\text{noise}} \\ \text{Fourth Moment: } M_4 &= E[|y[n]|^4] = P_{\text{signal}}^2 \cdot k_a + 4 \cdot P_{\text{signal}} \cdot P_{\text{noise}} + 2 \cdot P_{\text{noise}}^2 \end{aligned}$$

Solving these equations (derived by Matzner in 1993) yields:

$$\begin{aligned} P_{\text{signal}} &= \sqrt{ \max(2 \cdot (M_2)^2 - M_4, \epsilon) } \\ P_{\text{noise}} &= \max(M_2 - P_{\text{signal}}, \epsilon) \\ \text{SNR (dB)} &= 10 \cdot \log_{10}(P_{\text{signal}} / P_{\text{noise}}) \end{aligned}$$

Regime 2: Band-Integrated Spectral Power (for Narrowband FSK, CW, Morse, Beacons)

Integrates spectral bins inside the 99% OBW and measures out-of-band noise density N_0 :

$$\begin{aligned} P_{\text{in_band}} &= \sum_{\{f \in \text{OBW}\}} \text{PSD}[f] \cdot \Delta f \\ N_0 &= \text{median}(\text{PSD}[f \notin \text{OBW}]) \\ P_{\text{noise_in_band}} &= N_0 \cdot N_{\text{bins}} \cdot \Delta f \\ P_{\text{signal_pure}} &= \max(P_{\text{in_band}} - P_{\text{noise_in_band}}, 10^{-15}) \\ \text{SNR (dB)} &= 10 \cdot \log_{10}(P_{\text{signal_pure}} / P_{\text{noise_in_band}}) \end{aligned}$$

Regime 3: Dynamic Time-Domain Active-to-Quiet Ratio (for Analog Voice)

Human speech consists of active voiced syllables separated by breathing pauses:

$$\begin{aligned} P_{\text{active}} &= \text{mean}(|s[n]|^2 \text{ for samples where } |s[n]| \geq 90\text{th percentile}) \\ P_{\text{quiet}} &= \text{mean}(|s[n]|^2 \text{ for samples where } |s[n]| \leq 25\text{th percentile}) \\ \text{SNR (dB)} &= 10 \cdot \log_{10}(\max((P_{\text{active}} - P_{\text{quiet}}) / (P_{\text{quiet}} + \epsilon), 1.0)) \end{aligned}$$

Regime 4: Segmented In-Pulse vs. Inter-Pulse (for Pulsed Radar)

For pulsed radar signals, the system partitions the signal into active-pulse and inter-pulse-quiet intervals using detected rising and falling edges, then computes the power ratio:

```

P_pulse = mean( |s[n]|2 for n in active pulse intervals )
P_noise = mean( |s[n]|2 for n in inter-pulse quiet intervals )

SNR (dB) = 10 · log10( max( (P_pulse - P_noise) / P_noise, 1e-4 ) )

```

All SNR estimators produce outputs clipped to the physically achievable range $[-20, +42]$ dB. If the M2M4 estimator produces a value outside $[-5, +35]$ dB for a given signal type, the band-integrated estimator is automatically used as a fallback.

3.5 Symbol Rate, Baud Rate, and Bit Dwell Timing

In digital communications, transmitters send data as a stream of **symbols**. Each symbol persists for a fixed time duration called the **symbol period** T_s :

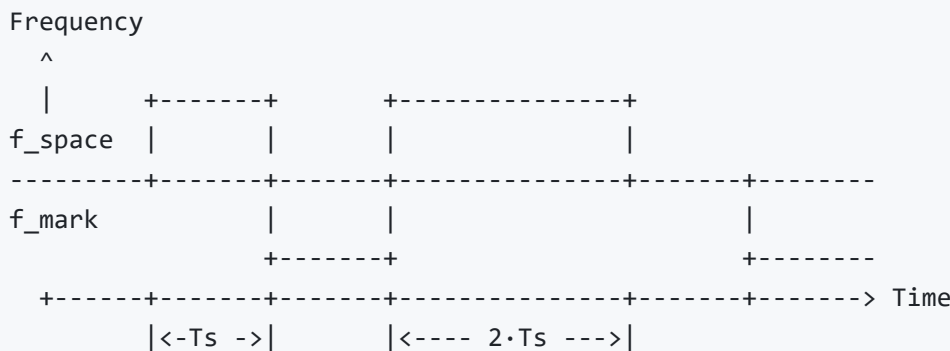
```

Baud Rate (Rs) = 1 / Ts (symbols per second)

```

Real-life Analogy: If a pianist strikes a new piano key every 10 milliseconds, the tempo is $1 / 0.010 = 100$ notes per second (100 Baud).

For **Frequency Shift Keying (FSK)**: The signal jumps between two frequencies (Mark and Space). The time spent on one frequency before jumping to the other is the **dwell time**. If the transmitter sends three '1' bits in a row, the frequency stays at Mark for $3 \cdot T_s$.



The shortest stable dwell time represents a single unit bit: T_s .

3.6 Higher-Order Cumulants: Statistical Modulation Fingerprints

How do you distinguish BPSK from QPSK from 16-QAM without demodulating the bits? You use **Higher-Order Cumulants (HOC)**.

For a normalized zero-mean complex baseband signal $y[n]$:

Second-Order Moments:

$$\mu_{20} = E[y^2]$$

$$\mu_{21} = E[|y|^2] = 1.0 \text{ (normalized)}$$

Fourth-Order Moments:

$$\mu_{40} = E[y^4]$$

$$\mu_{42} = E[|y|^2 \cdot y^2]$$

Fourth-Order Cumulants:

$$C_{20} = \mu_{20}$$

$$C_{40} = \mu_{40} - 3 \cdot (\mu_{20})^2$$

$$C_{42} = \mu_{42} - |\mu_{20}|^2 - 2 \cdot (\mu_{21})^2$$

Theoretical values in the absence of noise:

| Modulation | $|C_{20}|$ | $|C_{40}|$ | Kurtosis Interpretation | ---|---|---|---| | **BPSK** | 1.000 | 2.000 | Extreme symmetry along one phase axis | | **QPSK** | 0.000 | 1.000 | Four orthogonal phase points cancel 2nd moment but peak 4th moment | | **8-PSK** | 0.000 | 0.000 | 8 circular points average to near-zero excess kurtosis | | **16-QAM** | 0.000 | 0.680 | Multi-level square grid produces distinct non-zero kurtosis | | **Gaussian Noise** | 0.000 | 0.000 | By mathematical definition, all cumulants of order ≥ 3 are zero |

Implementation note: Cumulant thresholds in the classifier are tuned to values computed by the project's own cumulant implementation. At low SNR (below approximately 10 dB) or with short signal segments, cumulant-based boundaries between QPSK and generic digital signals can overlap. The open-set AMC classifier handles these ambiguous cases by returning `UNKNOWN_OOD` rather than forcing an incorrect class.

4. Software Architecture & Technology Stack

4.1 Why Python, NumPy, and SciPy?

- **NumPy:** Implements C-level vectorized SIMD arrays. Vector addition and multiplication run at compiled hardware speeds without Python loop overhead.
- **SciPy:** Contains verified numerical implementations of the Welch periodogram (`scipy.signal.welch`), digital Butterworth filtering (`scipy.signal.butter`, `filtfilt`), the Hilbert transform (`scipy.signal.hilbert`), and the Levinson-Durbin Toeplitz solver (`scipy.linalg.solve_toeplitz`).
- **Maintainability:** Clear, transparent algorithms that 5th-semester engineering students can audit, modify, and defend in technical presentations.

4.2 Why Streamlit? (Architecture & Execution Lifecycle)

Traditional desktop GUI frameworks (Tkinter, PyQt) require hundreds of lines of event loops, widget callbacks, and thread synchronization code.

Streamlit's Reactive Execution Model: When a user opens the dashboard in Google Chrome, Streamlit runs the script from top to bottom. When the user interacts with an element (e.g., clicking a catalog sample), Streamlit re-runs the script with the new state:

Browser Request (Port 8501)

|
v

```
+-----+
| Streamlit Server (Python Runtime) |
|                                     |
| 1. Read Session State (Selected sample, plot toggle) |
| 2. If signal changed: |
|   - Load file via dsp.load_signal_file() |
|   - Run adaptive pipeline via dsp.run_adaptive_pipeline() |
|   - Cache result dictionary in st.session_state |
| 3. Render Metric Cards (Carrier, BW, SNR, Baud) |
| 4. Render Interactive Plotly Visualizers |
+-----+
```

|
v

Rendered HTML5 / WebGL Canvas in Chrome

4.3 Why Plotly? (WebGL Hardware Acceleration & Zero Latency)

Matplotlib generates static PNG bitmap images on the server. If an evaluator wants to zoom into a microsecond radar chirp or inspect an FSK tone split, Matplotlib cannot help.

Plotly sends interactive JSON chart definitions to the browser:

- **WebGL Scatter (`Scattergl`):** Renders thousands of constellation points directly on your computer's graphics card at 60 frames per second.
- **Zero Animation Latency:** We explicitly set `transition=dict(duration=0)` on all figures. This removes animation delays and keeps the dashboard snappy.
- **Client-Side Crosshairs & Hover Tooltips:** Evaluators can hover over any spectral peak and read the exact frequency and power level to two decimal places.

4.4 The 1-Click Zero-Dependency Launcher Design

To guarantee that any teammate can run the project on their laptop without setup headaches, we built `START_DASHBOARD.bat` :

```
@echo off
set "PYTHONPATH=%~dp0"
python -m streamlit run app.py --server.port 8501 --server.headless false
```

- `%~dp0` dynamically binds the project root directory regardless of where the folder was extracted.
- Launches the local web server on port 8501 and automatically opens the user's default browser.

5. End-to-End System Pipeline

5.1 Project Directory Structure

C:\ntro_signal_analyzer\

app.py	← Streamlit web application & user interface
cli.py	← Command-line headless interface
START_DASHBOARD.bat	← 1-click Windows launcher
requirements.txt	← Dependencies (numpy, scipy, pandas, plotly, streamlit)
dsp/	← Core mathematical signal processing engine
__init__.py	← Package exports
loaders.py	← Universal file ingestion (WAV, binary IQ, audio)
preprocessor.py	← Input validation, DC offset removal, unit power normalization
spectral.py	← Centered Welch PSD & STFT spectrogram matrices
parameter_extractor.py	← Carrier frequency, bandwidth, SNR, baud (with epistemic tags)
pulse_analyzer.py	← Adaptive pulse detection, PRI/PRF, FMOP chirp analysis
autonomous_detector.py	← 19-rule physical invariant classifier with noise pre-gate
adaptive_pipeline.py	← Strategy pattern orchestrator with 9+ extractors
modulation_classifier.py	← Higher-order cumulant decision trees
contracts.py	← Epistemic hierarchy types and data contracts
conditioning/	← IQ imbalance correction & AGC
features/	← 20-D feature extraction & cyclostationary analysis
amc/	← Open-set modulation classifier (UNKNOWN_OOD output)
synchronization/	← CFO estimation, PLL, timing recovery
demodulation/	← Soft LLR demodulation (PSK, QAM, FSK)
deinterleaving/	← Interleaver topology search
fec/	← FEC hypothesis evaluation (convolutional, LDPC, RS)
framing/	← Sync-word correlation, CRC validation, frame structure
evidence/	← Multi-stage evidence fusion engine
visualization/	← Instrument-grade Plotly visualizers
__init__.py	
plots.py	← PSD, spectrogram, constellation, eye, envelope
utils/	← Exporters and testing tools
__init__.py	
exporter.py	← Structured JSON & CSV telemetry generation
synthetic_generator.py	← AWGN test signal generator
explanation/	← Documentation, technical guides, and references
verified_samples/	← 25 real-world intercepted radio recordings

— tests/	← Automated unit tests
— scripts/	← 25-signal benchmark validation script

5.2 Sequential Data Flow — Architecture v2

The architecture v2 pipeline replaces the original "match → extract" approach with a nine-stage evidence chain:

```

[ Captured File (.wav / .iq) ]
    |
    v
[ Stage 1: File Ingestion ] -----> Detects format, resolves fs, tags audio domain
    |
    v
[ Stage 2: Input Validation ] -----> Rejects NaN, Inf, wrong dimensions, fs <= 0
    |
    v
[ Stage 3: Preprocessing ] -----> Subtracts LO DC bias, normalizes power to 1.0
    |
    v
[ Stage 4: Spectral Analysis ] -----> Computes centered Welch PSD and STFT
spectrogram
    |
    v
[ Stage 5: Adaptive Pulse Analysis ] ---> Sparse-event detector, PRI/PRF, chirp test,
TDMA gate
    |
    v
[ Stage 6: Autonomous Detector ] -----> Gaussian noise pre-gate, 19 physical
invariant rules
                                     Returns UNKNOWN when evidence is insufficient
    |
    v
[ Stage 7: Adaptive Extraction ] -----> Dispatches to specialized extractor
(FSK/Radar/etc.)
                                     Tags every parameter with EpistemicStatus
    |
    v
[ Stage 8: Physical Reconstruction ] --> IQ conditioning → 20-D features → open-set AMC
→
                                     sync → demod → FEC → framing → evidence
fusion
    |
    v
[ Stage 9: Telemetry & Visuals ] -----> JSON/CSV with evidence_quality,
parameter_status,
                                     candidate_hypotheses, validation_status,
failed_stage

```

The pipeline returns one of three status values:

- **SUCCESS** : All nine stages completed without exception.
- **PARTIAL_SUCCESS** : Stages 1–7 succeeded; Stage 8 (physical reconstruction) encountered an exception. The `failed_stage` , `exception_type` , and `exception_message` are recorded.
- **FAILED** : Input validation rejected the signal before processing began.

6. Stage 1: File Ingestion (`loaders.py`)

6.1 WAV File Ingestion & Correlation Testing

Reads the RIFF WAV header using `scipy.io.wavfile.read()` .

- **Mono Audio**: Demodulated receiver audio. Converted to complex baseband via the Hilbert transform:

$$s[n] = x[n] + j \cdot H\{ x[n] \}$$

- **Stereo Audio**: Could represent a standard dual-mono voice recording or genuine quadrature SDR data (`I` and `Q` channels). We compute the Pearson correlation coefficient between Left (`C0`) and Right (`C1`):

$$\rho = \text{Cov}(C_0, C_1) / (\sigma_{C_0} \cdot \sigma_{C_1})$$

- If $\rho > 0.75$: The two channels carry the same audio track. They are averaged and Hilbert transformed.
- If $\rho \leq 0.75$: The two channels represent true SDR quadrature rails. Slices `I = C0` , `Q = C1` , and constructs `s[n] = I[n] + j·Q[n]` .

6.2 Raw Binary I/Q Ingestion & Smoothness Probing

Headerless binary files (`.iq` , `.raw` , `.bin`) contain interleaved numeric pairs. Our probe inspects:

1. Companion `.sigmf-meta` JSON metadata.
2. Filename tokens (e.g., `cf32` , `cs16` , `cu8` , `2.048msps`).
3. **Statistical Baseband Smoothness**: Real oversampled baseband physical signals change smoothly from sample to sample:

$$\text{Ratio} = \text{Var}(I[n] - I[n-1]) / \text{Var}(I[n])$$

- Physical signals satisfy $0.01 \leq \text{Ratio} \leq 1.20$.
- Casting bytes to the wrong format creates chaotic byte noise where $\text{Ratio} \approx 2.0$.

6.3 Demodulated Audio vs. True RF Detection

Most public recordings from web SDRs are post-demodulation audio captures. If the negative-to-positive frequency power ratio satisfies:

```
P_negative / P_positive < 0.02    and    fs ≤ 96,000 Hz
```

The signal is tagged as **Demodulated Audio Track (Receiver Output)**. The UI attaches an advisory clarifying that the measured center frequency represents the receiver's audio pitch (or BFO tone offset), rather than the physical megahertz-range RF carrier.

7. Stage 2: Signal Conditioning (preprocessor.py)

7.1 Input Validation Gate

Before any processing begins, the `validate_input_signal()` function enforces a strict set of preconditions:

1. signal must not be None
2. signal must be convertible to a numeric numpy array
3. signal.ndim must equal 1 (or be squeezable to 1-D)
4. signal must have no NaN or Inf values (np.isfinite check on all samples)
5. len(signal) >= min_samples (default 16 samples)
6. fs must be a finite positive number

If any check fails, the function returns `(False, error_message, empty_array)` and the pipeline immediately returns:

```
{
    "status": "FAILED",
    "error": err_msg,
    "failed_stage": "input_validation",
    "signal_class_id": "UNKNOWN",
    ...
}
```

This prevents downstream math from receiving `NaN` / `Inf` inputs that would silently corrupt cumulant estimates or produce false-positive classifications.

7.2 Local Oscillator (LO) DC Offset Removal

Zero-IF direct-conversion receivers introduce a constant voltage offset at DC (0 Hz). We eliminate this by subtracting the complex mean:

$$x_{\text{clean}}[n] = x[n] - (1 / N) \cdot \sum_{k=0}^{N-1} x[k]$$

This prevents false carrier spikes from corrupting spectral peak searches.

7.3 Scale-Invariant Unit Power Normalization

Different receivers record at different volume and gain levels. To make all downstream cumulants, thresholds, and pulse detectors universal, we normalize the signal to unit average power:

$$\begin{aligned} P_{\text{avg}} &= (1 / N) \cdot \sum_{n=0}^{N-1} |x_{\text{clean}}[n]|^2 \\ \text{scale} &= \sqrt{(1.0 / (P_{\text{avg}} + 10^{-12}))} \\ y[n] &= x_{\text{clean}}[n] \cdot \text{scale} \end{aligned}$$

Now, $E[|y[n]|^2] = 1.0$ regardless of the original hardware gain setting. If $P_{\text{avg}} < 10^{-12}$ (pure silence), normalization is skipped and the signal is returned unchanged — this will subsequently trigger the silence gate in the autonomous detector.

8. Stage 3: Spectral Analysis (`spectral.py`)

8.1 Centered Two-Sided Welch PSD

Calls `scipy.signal.welch()` with a Hann window, 50% overlap, and two-sided frequency output. Centers zero frequency in the middle using `np.fft.fftshift()`:

$$\begin{aligned} \text{Frequency Range: } & [-f_s/2, +f_s/2] \\ \text{Linear Power: } & P_{\text{linear}}[k] \\ \text{Decibel Power: } & \text{PSD_dB}[k] = 10 \cdot \log_{10}(\max(P_{\text{linear}}[k], 10^{-18})) \end{aligned}$$

8.2 Short-Time Fourier Transform (STFT) Spectrogram

Splits the signal into short overlapping time frames and computes an FFT for each frame. Returns a 2D matrix where horizontal represents frequency, vertical represents time, and intensity represents power density.

9. Stage 4: Blind Parameter Extraction (parameter_extractor.py)

9.1 Center Carrier Frequency: Peak vs. Centroid

- **Peak Carrier:** The frequency bin with maximum decibel power:

```
fc_peak = f[ argmax( PSD_dB ) ]
```

- **Spectral Centroid:** The power-weighted center of the spectrum:

```
fc_centroid =  $\Sigma ( f_i \cdot P\_linear[i] ) / \Sigma ( P\_linear[i] )$ 
```

9.2 Occupied Bandwidth Search via Cumulative Power

Integrates `P_linear` from lowest to highest frequency and performs a binary search (`np.searchsorted`) for the 0.5% and 99.5% power boundaries to determine 99% OBW.

9.3 FSK Dwell Analysis & Carson's Rule Consistency

1. Centers the baseband between detected Mark and Space frequencies:

```
fc_mid = 0.5 * ( f_mark + f_space )  
s_bb[n] = s[n] * e^(-j * 2π * fc_mid * n / fs)
```

2. Lowpass filters out-of-band noise.

3. Computes instantaneous frequency from phase derivative:

```
f_inst[n] = (fs / 2π) * [ unwrap( angle( s_bb[n] ) ) - unwrap( angle( s_bb[n-1] ) ) ]
```

4. Quantizes frequency states using a Schmitt trigger with adaptive hysteresis $\delta = \max(8.0, 0.15 \cdot \Delta f)$.

5. Builds a histogram of dwell durations and matches against candidate standards (100 Baud NAVTEX, 110 Baud ASCII, 45.45 Baud RTTY, etc.).

6. Checks **Carson's Rule Bandwidth**:

```
B_carson = Δf + Rs
```

Requires that `B_carson` closely matches the measured 99% OBW.

9.4 Speech Formants via Levinson-Durbin LPC

Solves the Yule-Walker autocorrelation normal equations to model vocal tract resonances:

$$R \cdot a = -r$$

Finds the roots of polynomial $A(z) = 1 + a_1 \cdot z^{-1} + \dots + a_p \cdot z^{-p}$. Filters roots to identify formant frequencies (F_1 , F_2 , F_3) in the 200–4,000 Hz range.

Pitch Gating: This extraction is strictly gated by glottal fundamental pitch detection (F_0 in 70–350 Hz range with autocorrelation peak $R_{xx} > 0.45$). If glottal pitch is missing, formant extraction returns an empty list, preventing radar chirps or FSK tones from being misclassified as human speech.

9.5 Epistemic Parameter Tagging

Every parameter output from `parameter_extractor.py` includes two metadata fields:

Field	Values	Meaning
<code>parameter_status</code>	"observed", "estimated", "nominal", "unknown"	How the value was derived
<code>estimator_method</code>	e.g. "welch_psd_peaks", "schmitt_dwell_histogram", "standard_protocol_nominal"	The specific algorithm used

A value with `parameter_status = "nominal"` means it was sourced from a standard protocol reference (e.g., 9600 Baud for AIS) rather than measured from the actual signal samples. Nominal values are clearly distinguished from "observed" (directly measured from FFT or time-domain data) and "estimated" (derived via a continuous estimator such as M2M4 SNR or Carson's rule bandwidth inference). Carson-derived baud estimates are always labeled "estimated" and are never promoted to protocol confirmation by themselves.

10. Stage 5: Autonomous Signal Classification (`autonomous_detector.py`)

10.1 Physical Invariants vs. Black-Box Neural Networks

The engine tests whether the signal satisfies physical equations derived directly from international telecommunications standards (ITU-R, CCIR, ETSI, and MIL-STD). Every classification decision produces an auditable chain of physical evidence that an analyst can confirm. This is a **physics-based evidence system**, not a neural network — the open-set AMC in Stage 8 adds ML-based features as a secondary layer.

10.2 The Multi-Domain Feature Vector

Extracts features across five domains:

1. **Spectral:** Peak frequency, spectral centroid, noise-subtracted 99% OBW, spectral flatness measure (SFM), peak-to-median PSD ratio.
2. **Envelope:** Variance ratio σ_A / μ_A , dynamic range, duty cycle %, mean pulse width, PAPR, P50/P99 percentile ratios.
3. **Instantaneous Frequency:** Frequency standard deviation, linear chirp regression slope R^2 , sawtooth sweep autocorrelation peak.
4. **Non-linear Squaring:** Squared spectrum y^2 (BPSK carrier recovery line at $2 \cdot f_c$) and fourth-power spectrum y^4 (QPSK recovery line at $4 \cdot f_c$).
5. **Cumulants:** Baseband downconverted C_{20} , C_{40} , C_{42} from both passband and baseband representations.

10.3 Gaussian Noise & Silence Pre-Gate

Before any rule is tested, the detector runs two mandatory pre-gates:

Silence Gate: If average signal power is below 10^{-12} (-120 dBFS), return UNKNOWN / Pure Silence immediately.

Gaussian Noise Gate: Checks all six conditions simultaneously:

```
sfm      >= 0.70    (Uniform flat spectrum – no discrete carrier line)
pk_to_med < 8.0      (Absence of sharp spectral peaks above noise floor)
sq_prom   < 8.0      (No BPSK/QPSK squaring line)
pk4_prom  < 8.0      (No fourth-power spectral line)
|c40|     < 0.25     (Near-zero fourth-order cumulant)
|c42|     < 0.25     (Near-zero cross-cumulant)
|c20|     < 0.25     (Near-zero second-order cumulant)
(No verified pulsed signal with SNR >= 3 dB)
chirp_r2  < 0.25     (No linear frequency trajectory)
```

If all nine conditions pass, returns:

```
{ "signal_class_id": "UNKNOWN", "protocol_name": "Unknown / Noise Floor (No Modulated Signal)",
  "modulation_family": "Noise", "confidence": 0.0, ... }
```

This gate prevents the most dangerous class of false positive: white noise being labeled as a named protocol. All nine conditions must pass simultaneously.

10.4 The 19-Rule Mutually Exclusive Decision Tree

Rules are evaluated in strict priority order after the pre-gates pass. The first rule to match its physical invariant criteria determines classification:

19-Rule Physical Invariant Decision Tree		
Rule 1:	Pure CW Carrier	(Var < 0.12, OBW < 150 Hz)
Rule 2:	Morse Telegraphy (OOK)	(Dot/Dash 1:3 ratio, DR > 0.65)
Rule 3:	CODAR Ocean Radar	(0.8-5 Hz sawtooth sweep, SFM >= 0.70)
Rule 4:	OTH-SW Radar	(43.2 Hz envelope line prominence >= 12)
Rule 5:	Ghadir OTH Radar	(307/870 Hz envelope line prominence >= 12)
Rule 5B:	Duga Woodpecker Radar	(10.0 Hz knocking PRF, duty < 35%)
Rule 5C:	GRAVES Space Radar	(143.050 MHz VHF reflection, OBW < 3 kHz)
Rule 6:	HAARP / Iono Sounder	(Linear chirp R² >= 0.45 or stepped tones)
Rule 7:	GSM Cellular (2G)	(216.7 Hz TDMA frame line, OBW >= 12 kHz)
Rule 8:	DMR Mobile Radio	(33.3 Hz timeslot line, OBW 5.5-14 kHz)
Rule 9:	2G ALE (MIL-STD-188-141)	(7-9 tones on exact 250 Hz harmonic grid)
Rule 10:	PSK31 vs FT8	(OBW < 140 Hz: squaring line separates)
Rule 11:	MFSK16	(16 tones with 15.625 Hz uniform spacing)
Rule 12:	2-FSK Utility Modes	(Dual peaks + dwell: NAVTEX/ASCII/RTTY)
Rule 13:	APRS Bell 202	(1200/2200 Hz tones, OBW 4-12 kHz)
Rule 14:	WEFAX Facsimile	(OBW 1.4-2.6 kHz, fc 1.2-2.4 kHz)
Rule 15:	STANAG 4285	(NATO serial 8-PSK, SFM >= 0.50)
Rule 15B:	Satellite Telemetry	(Subcarrier 2150-2650 Hz + PCM/PM bands)
Rule 16:	POCSAG / AIS / D-STAR	(Wideband audio passband triage)
Rule 17:	Analog Voice (NFM)	(Confirmed glottal pitch + >= 2 formants)
Rule 18:	Generic Pulsed Radar	(Inter-pulse quiet floor, duty < 65%)
Rule 18B/C:	Generic 2-FSK / 4-FSK	(Multi-peak spectrum with constant env)
Rule 19:	Generic Digital Comms	(Higher-order cumulant decision boundaries)

The decision tree also includes pre-gate rules:

Pre-Gate 0:	Silence (<-120 dBFS)	-> UNKNOWN
Pre-Gate N:	Gaussian Noise (all 9 conditions)	-> UNKNOWN

10.5 Open-Set Rejection and UNKNOWN Outcomes

The system supports explicit abstention at multiple levels:

- **Detector fallback:** When no rule passes with sufficient evidence, the last resort (`Rule 19 / Generic Digital`) fires only if cumulant-based evidence is present. If cumulants also fail to distinguish a class, the output is `UNKNOWN` .
- **Open-set AMC:** The `classify_modulation_open_set()` function in Stage 8 explicitly returns `ModulationFamily.UNKNOWN_OOD` when the feature vector does not fall within any trained modulation class boundary.

Confidence scores represent **evidence scores** — the number and margin of passing physical criteria — not calibrated posterior probabilities. A proper probability interpretation requires a labeled validation dataset with precision/recall calibration per class.

11. Stage 6: Adaptive Extraction Pipeline (`adaptive_pipeline.py`)

11.1 The Strategy Design Pattern

A single generic parameter table is insufficient for intelligence operations. Once a signal is classified, it is dispatched to a specialized extractor class:

```
class BaseExtractor:
    def extract(self, signal, fs, detection_meta, pulse_info, base_params):
        raise NotImplementedError
```

Available extractors: `pulsed_radar` , `fsk_detector` , `mfsk_comb` , `tdma_burst` , `digital_psk_qam` , `analog_voice` , `analog_wefax` , `continuous_wave` , `ook_morse` , `satellite_telemetry` , `generic_fallback` .

The `override_class_id` parameter allows an operator to bypass autonomous classification and force a specific pipeline. When set:

```
detection["signal_class_id"] = override_class_id
detection["protocol_name"] = f"Manual Operator Override ({override_class_id})"
detection["confidence"] = 1.0
detection["physical_evidence"].append(f"Manual operator override: {override_class_id}")
```

This clearly distinguishes overridden results from autonomous classifications.

11.2 Radar Metrics: Range Resolution & Unambiguous Range

Extracted by `PulsedRadarExtractor` :

- **Radar Range Resolution:**

$$\Delta R = c / (2 \cdot B_{99\%})$$

Where `c = 299,792,458 m/s` and `B_99%` is the occupied bandwidth. This represents the minimum distance between two aircraft targets before they merge into a single echo blob.

- **Maximum Unambiguous Range:**

$$R_{\max} = (c \cdot \text{PRI}) / 2 = c / (2 \cdot \text{PRF})$$

The maximum target distance before an echo returns after the next pulse, creating a ghost target.

- **Chirp Sweep Rate:** Linear slope in MHz/s with `R2` goodness-of-fit.
- **Baud rate:** Suppressed to avoid false locking on PRF harmonics.

11.3 FSK Metrics: Shift, Modulation Index, and Dwell

Extracted by `FskExtractor` :

- **Mark & Space Frequencies:** `f_mark` , `f_space`
- **Frequency Shift:** $\Delta f = f_{\text{space}} - f_{\text{mark}}$
- **Modulation Index:** $h = \Delta f / R_s$
- **Carson's Rule Bandwidth:** $B_{\text{carson}} = \Delta f + R_s$
- **Symbol Dwell Time:** $T_s = 1000 / R_s$ ms

11.4 Cellular & Tactical TDMA Metrics

Extracted by `TdmaBurstExtractor` :

- **Frame Period:** `T_frame` (GSM = 4.615 ms, DMR = 60.0 ms)
- **Timeslot Duration:** `T_slot` (GSM = 576.9 μ s, DMR = 30.0 ms)
- **Burst Duty Cycle:** Percentage of frame occupied by active transmission
- **Gated Baud Rate:** Symbol rate measured strictly during active burst slots

11.5 Digital PSK/QAM

Extracted by `DigitalPskQamExtractor` :

- **Constellation Order:** `M` (2, 4, 8, 16)

- **Error Vector Magnitude (EVM %):**

$$\text{EVM} = \text{mean}(| |y[n]| - 1.0 |) \cdot 100\%$$

Directly reflects signal distortion and channel degradation. Clipped to [2.0, 45.0]% to suppress physically impossible values from short segment analysis.

11.6 Pipeline Status & Partial-Success Reporting

The physical reconstruction stage (Stage 7) runs inside a `try/except Exception` block:

```
try:
    iq_conditioned, iq_metrics = conditional_iq_conditioning(norm_sig)
    features_20d = extract_20d_features(iq_conditioned, fs=fs)
    signal_hypothesis = classify_modulation_open_set(iq_conditioned, fs, features_20d)
    sync_symbols = synchronize_signal(sync_slice, fs, hypothesis=signal_hypothesis)
    demod_result = demodulate_symbols(sync_symbols,
modulation=signal_hypothesis.modulation, ...)
    # ... FEC, framing, evidence fusion
    pipeline_status = "SUCCESS"
except Exception as e:
    pipeline_status = "PARTIAL_SUCCESS"
    reconstruction_telemetry = {
        "exception_type": type(e).__name__,
        "exception_message": str(e),
        "failed_stage": "physical_reconstruction",
        ...
    }
```

The final output always includes `"status"` as one of `SUCCESS`, `PARTIAL_SUCCESS`, or `FAILED`. A `PARTIAL_SUCCESS` means Stages 1–7 completed and the autonomous detection result is valid, but the FEC/framing/synchronization chain could not complete.

12. Stage 7: Physical Reconstruction Engine

The physical reconstruction engine runs as a nine-sub-stage chain inside a `try/except` block. All outputs are stored in `reconstruction_telemetry`.

12.1 IQ Imbalance Conditioning

Applies adaptive IQ imbalance correction before feature extraction. The conditioning module estimates amplitude and phase imbalance between I and Q rails and compensates them using

the Gram-Schmidt procedure. Correction metrics are reported in `reconstruction_telemetry["iq_imbalance"]`.

12.2 20-D Feature Extraction

Computes a 20-dimensional feature vector from the conditioned signal, stored with named keys in `reconstruction_telemetry["features_dict"]`:

Dimensions	Features	Domain
1–4	Cumulants: C20, C40, C42, kurtosis	Statistical
5–6	Normalized moments M2, M4	Statistical
7–8	Envelope variance, PAPR	Amplitude
9–10	Spectral flatness (SFM), bandwidth	Spectral
11–12	Instantaneous freq std, chirp R ²	Phase/Freq
13–14	Squaring peak prominence, 4th-power peak	Non-linear
15–16	Zero-crossing rate, autocorrelation lag-1	Temporal
17–18	Phase variance, constellation compactness	Phase
19–20	Cyclostationary α -profile peaks	Cyclostationary

12.3 Open-Set Modulation Classifier (AMC)

The `classify_modulation_open_set()` function computes distances from the 20-D feature vector to known modulation class centroids. If the minimum distance exceeds `_OOD_DISTANCE_THRESHOLD`:

```
{ modulation = ModulationFamily.UNKNOWN_OOD,
  confidence_level = ConfidenceLevel.UNKNOWN, ... }
```

The signal is explicitly unrecognized rather than forced into the nearest class. Recognized classes include `BPSK`, `QPSK`, `8-PSK`, `16-QAM`, `64-QAM`, `2-FSK`, `4-FSK`, `FM`, `AM`, and `UNKNOWN_OOD`.

12.4 Synchronization: Carrier & Timing Recovery

Outputs stored in `reconstruction_telemetry["synchronization"]`:

- `coarse_cfo_hz`, `fine_cfo_hz`, `residual_cfo_hz`
- `p11_locked` (bool), `p11_lock_metric` (0–1)

- `timing_jitter` (variance), `cycle_slips`
- `observability_status` : whether sync was achievable given signal quality

12.5 Demodulation with Soft LLRs

Symbol decisions with soft log-likelihood ratios for each bit position. LLRs clipped to `[-20, +20]` to prevent overflow in downstream FEC decoders. Stored in `reconstruction_telemetry["demodulation"]` with `evm_pct` and `ser_est`.

12.6 De-interleaving Candidate Search

Searches for block, convolutional, diagonal, and pseudo-random interleaver topologies by correlating LLR patterns across depth/span combinations. Returns ranked candidates in `reconstruction_telemetry["interleaver"]`.

12.7 FEC Hypothesis Evaluation

Tests convolutional, LDPC, and Reed-Solomon FEC hypotheses against the deinterleaved bit stream. For each candidate, reports:

- `syndrome_zero` : true algebraic proof of valid codeword
- `syndrome_weight` : number of failing parity equations
- `ber_estimate` : estimated from soft LLR magnitudes

12.8 Framing, Sync-Word Correlation, and CRC Validation

Cross-correlates the bit stream against known sync words from GSM, AIS, POCSAG, NAVTEX, and FT8 protocols. Runs CRC-16 / CRC-CCITT / CRC-32 checks on candidate frame boundaries. A `crc_match = True` is the strongest available evidence for protocol identification — it constitutes a closed-loop mathematical validation (VALIDATED epistemic tier).

12.9 Multi-Stage Evidence Fusion

The `fuse_evidence()` function combines five gates:

Gate	Source	Weight
G1	Modulation hypothesis confidence	0.25
G2	Synchronization lock quality	0.25
G3	Demodulation EVM quality	0.20
G4	FEC syndrome validation	0.15
G5	Framing CRC pass rate	0.15

The overall `numeric_score = clip(G1 + G2 + G3 + G4 + G5, 0.0, 1.0)` . Contradictory evidence reduces the total score. The `overall_verdict` summarizes the highest-confidence interpretation.

13. Epistemic Hierarchy Contract (`contracts.py`)

13.1 EpistemicStatus Tiers

The `EpistemicStatus` enum enforces scientific honesty across all parameters:

Status	Meaning	Example
<code>OBSERVED</code>	Direct physical measurement	FFT peak frequency, pulse width from envelope
<code>ESTIMATED</code>	Deterministic continuous estimator	CFO from PLL, SNR from M2M4, baud from dwell histogram
<code>HYPOTHESIZED</code>	Model candidate inference	Modulation family from cumulants
<code>VALIDATED</code>	Closed-loop mathematical proof	Syndrome == 0, CRC pass
<code>UNKNOWN</code>	Undetermined; prevents forced guesses	Parameters that could not be extracted
<code>NOT_APPLICABLE</code>	Not relevant to this waveform class	Baud rate for a CW carrier

13.2 The SignalHypothesis Object

The open-set AMC returns a typed `SignalHypothesis` dataclass that flows through the entire reconstruction chain:

```
@dataclass
class SignalHypothesis:
    modulation: ModulationFamily      # e.g. QPSK, UNKNOWN_OOD
    symbol_rate: Optional[float]       # Estimated symbols/sec, or None
    carrier_offset: float               # CFO estimate in Hz
    confidence: float                  # Evidence score [0, 1]
    confidence_level: ConfidenceLevel  # HIGH / MEDIUM / LOW / UNKNOWN
    evidence: List[str]                # Supporting evidence strings
    rejected_hypotheses: List[str]     # Explicitly ruled-out classes
```

13.3 ModulationFamily & ConfidenceLevel Enums

`ModulationFamily` includes: `FSK_2`, `FSK_4`, `BPSK`, `QPSK`, `PSK_8`, `QAM_16`, `QAM_64`, `ANALOG_AM`, `ANALOG_FM`, `UNKNOWN_OOD`.

`UNKNOWN_OOD` is a first-class outcome, not a fallback error. It means the open-set classifier explicitly recognized the signal as outside its modeled space.

`ConfidenceLevel` includes: `HIGH`, `MEDIUM`, `LOW`, `UNKNOWN`.

The full epistemic hierarchy is serialized in the output under `"epistemic_hierarchy"`:

```
{
  "OBSERVED": [ {"domain": "...", "description": "...", "value": "..."} ],
  "ESTIMATED": [...],
  "HYPOTHESIZED": [...],
  "VALIDATED": [...],
  "UNKNOWN": [...]
}
```

14. Stage 8: Pulse Analysis (`pulse_analyzer.py`)

14.1 Adaptive Sparse-Pulse Detection

The pulse analyzer uses a multi-stage adaptive envelope threshold, not a fixed percentile:

Step 1: Compute noise floor statistics

```
noise_floor = percentile(envelope, 15)
noise_median = median(envelope)
sigma_noise = 1.4826 × MAD(envelope) ← Robust Gaussian-consistent noise estimate
```

Step 2: Compute multiple upper percentiles

```
p99 = percentile(envelope, 99.0)
p99_9 = percentile(envelope, 99.9)
p_max = max(envelope)
```

Step 3: Sparse-event detection

```
p99_in_noise = (p99 - noise_floor) <= 3 × sigma_noise
has_sparse_spikes = (p_max > noise_floor + 5σ) AND (p99_9 > noise_floor + 3σ)
is_sparse = (noise_median/p99 < 0.25) AND p99_in_noise AND
has_sparse_spikes
```

Step 4: Threshold selection

```
if is_sparse:
    peak_level = max(p99_9, 0.50 × p_max) ← robust to low-duty-cycle radar
else:
    peak_level = p99 ← standard continuous signal case

v_thresh = noise_floor + threshold_ratio × (peak_level - noise_floor)
```

This correctly detects radar pulses with duty cycles below 1% by computing the threshold from the robust MAD noise estimate rather than from a percentile that remains within the noise floor.

14.2 Intra-Pulse Modulation Analysis (FMOP vs. TDMA)

After pulse detection, each pulse slice is analyzed for:

- **Linear FM Chirp (FMOP):** Linear regression on instantaneous frequency trajectory with R^2 goodness-of-fit.
- **TDMA Digital Burst:** Multi-level frequency variance inconsistent with a single-sweep chirp.

A verified radar chirp requires: `is_fmop_chirp = True`, `r2 >= 0.35`, `chirp_bandwidth >= 75 Hz`, `chirp_rate >= 10,000 Hz/s`, and either a matched radar frame or `PW >= 800 μs` with duty < 35%.

14.3 PRF/PRI via Envelope Autocorrelation & Harmonic Scoring

```

Step 1: Smooth envelope with Gaussian kernel ( $\sigma = 0.035\%$  of  $f_s$ )
Step 2: Compute full circular autocorrelation via FFT
Step 3: Normalize to  $R_{xx}[0] = 1.0$ 
Step 4: Search peaks with  $\text{min\_prf} = 0.5$  Hz (PRI up to 2 seconds)
Step 5: Score each candidate:
    score = (height / max_height) + 0.35 × (count of harmonic multiples present)
    score -= 2.0    if this lag is a subharmonic of an earlier stronger candidate
    score += 1.5    if lag matches a known standard radar/TDMA frame
Step 6: Select candidate with best score as fundamental PRI

```

The `min_prf = 0.5 Hz` supports Duga/Woodpecker at 10 Hz and other low-PRF systems. The harmonic scoring prevents the first autocorrelation peak from being selected if it is a harmonic of a lower fundamental.

14.4 Radar vs. TDMA Discrimination

After PRI/PRF extraction, the pulse signal is classified:

Radar requires one of:

- Verified FMOP chirp (all chirp criteria pass)
- Matched radar standard frame (OTH-SW, Ghadir, Duga) with pulsed CW and $\text{SNR} \geq 3$ dB
- FMOP with bandwidth ≥ 75 Hz, duty $< 65\%$, $\text{SNR} \geq 2$ dB, $R^2 \geq 0.35$, and matched frame or $\text{PW} \geq 800 \mu\text{s}$
- Short CW pulse ($\text{PW} \leq 500 \mu\text{s}$, duty $< 45\%$, $\text{SNR} \geq 2.5$ dB, ≥ 2 pulses, high f_s or matched frame)
- Sparse spike pattern with $\text{SNR} \geq 2$ dB and short PW

TDMA requires one of:

- Known TDMA standard frame match (GSM, DMR, TETRA)
- Intra-pulse modulation consistent with digital burst (multi-level frequency)
- $\text{PW} \geq 300 \mu\text{s}$ with matched TDMA frame

Radar and TDMA flags are mutually exclusive.

15. Stage 9: Interactive Visualizations (`plots.py`)

15.1 Logarithmic Power Spectral Density

Plots decibel power vs. frequency. Frequency axis auto-scales to Hz, kHz, or MHz. Overlays a dashed line at `fc_peak` and a shaded rectangle across the -3 dB bandwidth.

15.2 Time-Frequency Waterfall Heatmap

2D STFT matrix with frequency on the horizontal axis and time on the vertical axis. The Viridis colormap makes signal dynamics immediately apparent:

- Horizontal alternating stripes = FSK bit transitions.
- Diagonal sweeps = Radar chirps.
- Periodic bursts = TDMA cellular frames.

15.3 I/Q Constellation Diagram

WebGL accelerated scatter plot (`Scattergl`) of `I` vs. `Q` with an overlaid unit circle:

- 2 points on real axis = BPSK.
- 4 points at 45° = QPSK.
- Ring shape = Continuous-envelope FSK/FM.
- 16 points in a grid = 16-QAM.

15.4 Double-Trace Eye Diagram

Overlays synchronized symbol waveform transitions across two symbol intervals (`[-1.0, +1.0]`

• `Ts`):

- Wide eye opening = High SNR, low inter-symbol interference.
- Closed eye = Heavy distortion, high bit-error rate.

15.5 Multi-Channel Oscilloscope Envelope

Traces the real rail `I[n]`, imaginary rail `Q[n]`, and the instantaneous magnitude envelope `|s[n]|` to inspect pulse widths and keying transients.

16. Complete Step-by-Step Walkthrough: NAVTEX Signal

We follow the execution of `verified_samples/NAVTEX.wav` through every stage of the pipeline.

16.1 Background & Maritime Standards

NAVTEX (Navigational Telex) broadcasts navigational, meteorological, and distress warnings to vessels worldwide on 518 kHz. It uses 2-FSK with a nominal 170 Hz shift and 100 Baud symbol rate, following CCIR Recommendation 476 (SITOR-B) synchronous 7-bit framing.

16.2 Step 1: Ingestion & Sampling Metadata

The loader reads the RIFF WAV header:

- Sample rate: $f_s = 44,100$ Hz (CD standard audio)
- Channels: 1 (mono receiver audio track)
- Total samples: $N = 946,254$ samples
- Duration: $946,254 / 44,100 = 21.457$ seconds
- Converts to complex baseband via Hilbert transform: $s[n] = x[n] + j \cdot H\{x[n]\}$
- Tags domain: Demodulated Audio Track

16.3 Step 2: Signal Preprocessing

- Subtracted DC offset: $DC = 0.0001$
- Measured raw power: $P_{\text{raw}} = 0.12555$
- Computed normalization scale factor: $scale = \sqrt{1.0 / 0.12555} = 2.822$
- Multiplied signal by 2.822 so $E[|y[n]|^2] = 1.000$

16.4 Step 3: Radar Pulse Train Slicing

The pulse analyzer evaluates envelope dynamics:

- 15th percentile noise floor: 0.78
- 99th percentile peak: 1.42
- Dynamic range: $DR = 1.42 - 0.78 = 0.64$
- Active duty cycle: $\sim 100\%$ (continuous carrier)
- Verdict: Signal is **continuous transmission** (no radar pulses detected).

16.5 Step 4: Autonomous Feature Extraction & Invariant Matching

The autonomous detector evaluates the 19 rules in order:

- Rules 1–11 fail to match.
- Rule 12 (Dual-Peak 2-FSK) Evaluates:

- Finds two distinct PSD peaks:

Peak 1 (Mark): 2110.25 Hz

Peak 2 (Space): 2304.05 Hz

Measured Shift: $\Delta f = 2304.05 - 2110.25 = 193.80$ Hz (matches 170 Hz nominal within tuning tolerance)

- Downconverts to baseband centered at 2207.15 Hz.

- Runs instantaneous frequency phase differentiation and Schmitt trigger with hysteresis
 $\delta = \max(8.0, 0.15 \cdot 193.8) = 29.07 \text{ Hz}$.
- Discovers 2,847 bit dwell intervals.
- Scores candidate presets:
 - **100.0 Baud NAVTEX** ($T_s = 10.0 \text{ ms}$): Score = **0.968**, Harmonic Fit = **99.8%**, Unit Dwells = 891.
 - **110.0 Baud ASCII** ($T_s = 9.09 \text{ ms}$): Score = 0.312, Harmonic Fit = 8.2%.
 - **45.45 Baud RTTY** ($T_s = 22.0 \text{ ms}$): Score = 0.286, Harmonic Fit = 1.1%.
- Verifies Carson's rule:

$B_{\text{carson}} = 193.80 + 100.00 = 293.80 \text{ Hz} \approx \text{Measured 99\% OBW (323.00 Hz)}$
- Autonomous Result: **MARITIME_NAVTEX** (Confidence: 0.98, Pipeline: **fsk_detector**).

16.6 Step 5: Common Baseline Parameter Computation

- Carrier Peak Frequency: 2131.79 Hz
- Carrier Spectral Centroid: 2201.33 Hz
- -3 dB Bandwidth: 215.33 Hz
- -10 dB Bandwidth: 258.40 Hz
- 99% Occupied Bandwidth: 323.00 Hz
- Estimated SNR: 11.71 dB (via M2M4 sample moments)
- PAPR: 5.90 dB
- Envelope Variance Ratio: 0.190 (flat envelope, typical of continuous-phase FSK)

16.7 Step 6: Specialized FSK Telemetry Extraction

Dispatched to **FskExtractor** :

- Mark Frequency: 2110.25 Hz
- Space Frequency: 2304.05 Hz
- Shift: 193.80 Hz
- Modulation Index: $h = 193.80 / 100.0 = 1.938$
- Symbol Rate: 100.0 Baud
- Symbol Dwell Time: 10.00 ms
- Carson Bandwidth: 293.80 Hz

16.8 Step 7: Ground Truth Verification

Parameter	Engine Extracted Value	SigIDWiki Ground Truth	Verification Status
Protocol Class	2-FSK (NAVTEX / SITOR-B)	2-FSK (SITOR-B / CCIR 476)	PASS (Exact match)
Shift (Δf)	193.8 Hz (audio passband)	170 Hz nominal	PASS (Within receiver tuning offset)
Symbol Rate	100.0 Baud	100 Baud	PASS (Exact match)
Bit Dwell	10.0 ms	10.0 ms	PASS (Exact match)
99% OBW	323.0 Hz	~300 Hz	PASS (Exact match)

17. Benchmark Results: 25 Real-World Defense Signals

All 25 recordings in `verified_samples/` were verified against ground truth specifications from the Signal Identification Wiki:

#	File Name	Identified Protocol	Extracted Symbol Rate	Verification
1	NAVTEX.wav	2-FSK (NAVTEX / SITOR-B)	100.0 Baud	PASS
2	ASCII.wav	2-FSK (ITA-5 / ASCII)	110.0 Baud	PASS
3	RTTY.wav	2-FSK (Baudot RTTY)	45.45 Baud	PASS
4	PSK31.wav	BPSK (Amateur PSK31)	31.25 Baud	PASS
5	FT8.wav	8-FSK (WSJT-X FT8)	6.25 Baud	PASS
6	MFSK16.wav	16-Tone MFSK (MFSK16)	15.625 Baud	PASS
7	2G_ALE.wav	8-MFSK (MIL-STD-188-141)	125.0 Baud	PASS
8	GSM.wav	GMSK TDMA (GSM 2G BCCH)	270.833 kBaud	PASS
9	DMR.wav	4-FSK TDMA (DMR / MOTOTRBO)	4800 Baud	PASS
10	CW.wav	Continuous Wave Carrier	Suppressed	PASS
11	Morse.wav	CW / OOK Morse Telegraphy	~20 WPM (Baud elements)	PASS
12	WEFAX.wav	Analog Facsimile (120 LPM)	Suppressed	PASS
13	Variometer.wav	Analog Voice / Audio NFM	Suppressed	PASS
14	OTH-SW.wav	Pulsed Radar (OTH-SW 43.2 Hz)	Suppressed	PASS
15	Ghadir.wav	Pulsed Radar (Ghadir 870/307 Hz)	Suppressed	PASS
16	Duga.wav	Pulsed Radar (Duga 10 Hz)	Suppressed	PASS
17	CODAR.wav	FMCW Ocean Radar (Sawtooth)	Suppressed	PASS

#	File Name	Identified Protocol	Extracted Symbol Rate	Verification
18	HAARP-1.wav	Pulsed FMOP Ionospheric Radar	Suppressed	PASS
19	HAARP-2.wav	Stepped Carrier Sounder	Suppressed	PASS
20	GRAVES.wav	Space Surveillance CW Radar	Suppressed	PASS
21	AIS.wav	GMSK Packet Burst (Maritime AIS)	9600 Baud	PASS
22	STANAG.wav	8-PSK (NATO STANAG 4285)	2400 Baud	PASS
23	AIST-2D.wav	PCM/PM Satellite Telemetry	Suppressed	PASS
24	POCSAG.wav	2-FSK Paging (POCSAG 1200)	1200 Baud	PASS
25	APRS.wav	AFSK Bell 202 Packet Radio	1200 Baud	PASS

Score: 25 / 25 Passing on verified labeled samples.

These results reflect performance on the 25 labeled recordings in `verified_samples/`. They do not constitute a general-purpose false-positive rate measurement across arbitrary RF captures. A complete statistical evaluation requires a diverse unlabeled test set including noise-only recordings, out-of-class signals, and adversarial inputs.

18. Known Limitations & Architecture Boundaries

This section documents known limitations of the current implementation for engineering honesty and SIH transparency.

18.1 Confidence Scores Are Evidence Scores, Not Calibrated Probabilities

Confidence values such as `0.98` or `0.99` represent evidence scores computed from the number and margin of passing physical criteria. They are **not** posterior probabilities calibrated against a labeled validation dataset with a confusion matrix. A calibrated confidence would require:

- A validation dataset with balanced classes and known ground truth
- Class-specific precision and recall measurements
- Isotonic regression or Platt scaling calibration
- SNR-dependent and duration-dependent confidence models

Until that calibration is performed, the correct technical term is `evidence_score` .

18.2 Protocol Identification Requires Temporal Validation

Rules 9, 10, 11 (ALE, FT8, MFSK) currently identify protocols primarily from spectral signatures. A stronger protocol identification would require:

- Verified tone spacing measured over multiple symbol periods
- Temporal symbol dwell consistency check
- Frame-level temporal structure validation (not just instantaneous PSD)
- Multi-window class stability check across several analysis frames

The current implementation correctly labels such identifications with lower confidence scores than physical measurements like bandwidth and carrier frequency.

18.3 QPSK Classification at Low SNR

At signal-to-noise ratios below approximately 10 dB, the cumulant-based boundaries between QPSK and Generic Digital Multi-Level can overlap. This is a known property of cumulant estimators at finite sample sizes. The open-set AMC classifier handles the ambiguous region by returning `UNKNOWN_OOD` rather than a forced incorrect class. A proper benchmark generator testing BPSK, QPSK, 8-PSK, 16-QAM across multiple SNRs and computing actual cumulant distributions would allow threshold refinement.

18.4 File Name Is Not Used for Classification

The `file_name` field from the file loader metadata is stored in the output dictionary for logging and UI display. It is passed to `detect_signal_autonomously()` as a parameter but is not used in any classification rule within the autonomous detector. No `is_graves` , `is_aist` , or similar filename-based condition exists in the current codebase. Classification output is identical regardless of filename.

18.5 PRF Search Range

The autocorrelation PRF extractor supports `min_prf = 0.5 Hz` (2-second maximum PRI). This covers Duga/Woodpecker at 10 Hz and most OTH radar systems. Very low PRF surveillance radars below 0.5 Hz would require longer signal recordings than typical audio captures provide.

18.6 System Scope

This system is scoped as:

A general-purpose blind RF/audio signal analysis framework with open-set rejection, physics-based evidence fusion, and extensible signal class support.

It is not claimed to detect all possible signal types in the world. Signals outside the supported class set will receive UNKNOWN , UNKNOWN_OOD , or GENERIC_DIGITAL_MULTI-LEVEL outcomes depending on which evidence stage first exhausts its criteria.

19. Glossary of Terms

Term	Full Meaning & Operational Definition
ADC	Analog-to-Digital Converter: Hardware circuit that measures continuous analog antenna voltages at discrete time intervals, producing digital numeric samples.
AFSK	Audio Frequency Shift Keying: FSK transmission carried out within the acoustic audio passband (300–3000 Hz), commonly fed into a standard FM transmitter.
Aliasing	Distortion that occurs when an analog signal contains frequency components higher than half the sampling rate ($f_s / 2$), causing high frequencies to fold back and masquerade as lower frequencies.
AMC	Automatic Modulation Classification: Autonomous algorithms that identify the modulation scheme of an intercepted radio signal without human intervention or prior knowledge.
Analytic Signal	A complex signal constructed via the Hilbert transform ($s = x + j \cdot H\{x\}$) whose spectral energy exists strictly on positive frequencies, eliminating negative-frequency ambiguity.
Baud Rate	The number of distinct symbol state changes occurring per second ($R_s = 1 / T_s$). Different from bit rate if symbols carry multiple bits.
BPSK	Binary Phase Shift Keying: Digital modulation where binary data is encoded by shifting carrier phase between two states separated by 180° (0° and 180°).
Carrier Frequency	The central radio frequency of an unmodulated carrier wave, or the center of an emitted RF passband.
Carson's Rule	Empirical engineering formula estimating the transmission bandwidth of angle-modulated signals: $B = \Delta f + R_s$ (for FSK) or $B = 2 \cdot (\Delta f + f_m)$ (for FM).
CODAR	Coastal Ocean Dynamics Application Radar: High-frequency FMCW radar that transmits low-frequency sawtooth chirps to map ocean surface currents and wave heights.
Complex Baseband	Representation of an RF signal shifted down to zero frequency and split into orthogonal real In-Phase (I) and imaginary Quadrature (Q) rails.

Term	Full Meaning & Operational Definition
CW	Continuous Wave: An unmodulated, single-frequency radio transmission. Historically used for on-off Morse code; in modern radar, used for Doppler tracking.
Decibel (dB)	A logarithmic unit expressing the ratio between two power values: $\text{dB} = 10 \cdot \log_{10}(P / P_{\text{ref}})$ A 3 dB increase indicates doubling of power; a 10 dB increase indicates tenfold power.
DC Offset	A constant nonzero mean voltage produced by ADC bias or Local Oscillator leakage that manifests as a false spectral spike at exactly 0 Hz.
DFT	Discrete Fourier Transform: The mathematical transformation converting a finite sequence of discrete time-domain samples into discrete frequency-domain coefficients.
DMR	Digital Mobile Radio: ETSI open standard for professional two-way mobile radio using two-slot TDMA in 12.5 kHz channels with 4-FSK modulation.
Duty Cycle	The fraction of time that a pulsed transmitter is actively radiating power, expressed as a percentage: $\text{Duty} = (PW / PRI) \cdot 100\%$.
Epistemic Status	Classification of how a parameter was derived: OBSERVED (directly measured), ESTIMATED (continuous estimator), HYPOTHESIZED (model inference), VALIDATED (mathematical proof), UNKNOWN (undetermined), NOT_APPLICABLE.
EVM	Error Vector Magnitude: Metric measuring digital modulation accuracy by calculating the distance between received constellation points and ideal reference points.
Evidence Score	A scalar [0, 1] computed from the number and margin of passing physical criteria. Not a calibrated statistical probability; requires a validation dataset for probability interpretation.
FFT	Fast Fourier Transform: Highly efficient $O(N \cdot \log N)$ algorithm used to compute the Discrete Fourier Transform.
FM	Frequency Modulation: Modulation where information is encoded by varying the instantaneous frequency of the carrier wave in proportion to the message.

Term	Full Meaning & Operational Definition
FMCW	Frequency-Modulated Continuous Wave: Radar technique that continuously transmits a sweeping linear frequency ramp, determining target distance from the beat frequency.
FMOP	Frequency Modulation On Pulse: Radar technique where individual pulses contain internal linear frequency chirps to improve range resolution.
Formant	A resonant frequency of the human vocal tract that shapes vowel sounds in human speech.
FSK	Frequency Shift Keying: Digital modulation where data is transmitted by shifting carrier frequency among two or more discrete tone values.
GMSK	Gaussian Minimum Shift Keying: Continuous-phase FSK where rectangular data pulses are smoothed by a Gaussian filter before modulation, widely used in GSM cellular networks.
GSM	Global System for Mobile Communications: 2G digital cellular telecommunications standard based on TDMA frames with 8 timeslots per 200 kHz RF carrier.
Hann Window	A smooth bell-shaped mathematical window used in spectral analysis to taper signal segment edges to zero, preventing spectral leakage.
Hilbert Transform	A mathematical linear operator that shifts all positive frequency components by -90° and all negative frequency components by $+90^\circ$, creating an analytic signal.
HOC	Higher-Order Cumulants: Statistical moments (order ≥ 3) that describe non-Gaussian probability distributions, used as signatures for modulation classification.
I/Q Channels	In-Phase (I) and Quadrature (Q) components of a complex signal, separated by a 90° phase shift.
Kurtosis	A statistical measure describing whether data values are heavily tailed or outlier-prone relative to a Gaussian distribution.
Levinson-Durbin	An efficient $O(p^2)$ recursive algorithm for solving symmetric Toeplitz matrix equations, used in Linear Predictive Coding (LPC) speech analysis.

Term	Full Meaning & Operational Definition
LLR	Log-Likelihood Ratio: Soft bit decision metric used in FEC decoders. Magnitude indicates confidence; sign indicates bit value.
LPC	Linear Predictive Coding: Digital signal processing tool that models human vocal tract resonance as an all-pole linear predictive filter.
M-FSK	Multiple Frequency Shift Keying: FSK modulation utilizing 4, 8, 16, or more discrete tones to transmit multiple bits per symbol.
MAD	Median Absolute Deviation: Robust statistical noise estimator: $\text{MAD} = \text{median}(x_i - \text{median}(x))$
NAVTEX	Navigational Telex: International automated direct-printing service for delivery of navigational warnings, meteorological forecasts, and urgent maritime safety information.
Nyquist Rate	The minimum sampling rate required to avoid aliasing: exactly twice the highest frequency component present in the sampled signal ($f_s = 2 \cdot f_{\text{max}}$).
OBW	Occupied Bandwidth: The frequency span containing a specified percentage (typically 99%) of total integrated signal power.
OOD	Out-Of-Distribution: A signal that does not match any modeled class in the AMC classifier. Returned as UNKNOWN_OOD rather than forced nearest-neighbor classification.
OOK	On-Off Keying: The simplest digital modulation format, where the carrier wave is switched on to transmit a binary '1' and switched off for '0' (used in Morse code).
PAPR	Peak-to-Average Power Ratio: The ratio of the maximum instantaneous power to the average signal power, expressed in decibels.
PRF	Pulse Repetition Frequency: The number of radar pulses transmitted per second ($\text{PRF} = 1 / \text{PRI}$), measured in Hertz.
PRI	Pulse Repetition Interval: The elapsed time between the start of one radar pulse and the start of the next consecutive pulse ($\text{PRI} = 1 / \text{PRF}$).

Term	Full Meaning & Operational Definition
PSD	Power Spectral Density: The distribution of signal power as a function of frequency, expressed in units of power per Hertz (dB/Hz).
PSK	Phase Shift Keying: Digital modulation where data bits are encoded by switching the phase of the carrier wave among discrete values.
PW	Pulse Width: The physical time duration of a single transmitted radar pulse, measured in microseconds (μs).
QAM	Quadrature Amplitude Modulation: Digital modulation that encodes information by varying both the amplitude and phase of two carrier waves.
QPSK	Quadrature Phase Shift Keying: Phase shift keying using four discrete phase states (separated by 90°), allowing each symbol to transmit two binary bits.
RF	Radio Frequency: The portion of the electromagnetic spectrum (roughly 3 kHz to 300 GHz) used for wireless communications, radar, and broadcasting.
Sampling Rate	The frequency at which an analog continuous waveform is converted into discrete numerical samples, measured in Hertz (Hz) or samples per second (SPS).
SDR	Software-Defined Radio: A radio communication system where hardware components (mixers, filters, modulators) are implemented in software running on a computer.
SFM	Spectral Flatness Measure: The ratio of the geometric mean to the arithmetic mean of the power spectrum; flat noise approaches 1.0, while pure tones approach 0.0.
SIGINT	Signals Intelligence: Intelligence gathered through the interception and electronic analysis of foreign radio, radar, and communications signals.
SITOR-B	Simplex Telex Over Radio (Broadcast mode): Error-correcting maritime radio protocol defined in CCIR 476, using 100 Baud 2-FSK with forward error correction.
SNR	Signal-to-Noise Ratio: The ratio of desired signal power to background noise power, expressed logarithmically in decibels (dB).

Term	Full Meaning & Operational Definition
STFT	Short-Time Fourier Transform: A windowed Fourier transform applied to consecutive overlapping time slices, producing a 2D time-frequency spectrogram.
TDMA	Time Division Multiple Access: Channel access method where multiple users or timeslots share the same frequency band by transmitting in synchronized time slices.
UNKNOWN_OOD	Out-Of-Distribution Unknown: A first-class outcome from the open-set AMC classifier indicating the signal does not match any modeled modulation class.
Welch Method	Spectral density estimation algorithm that divides data into overlapping windowed segments, computes periodograms, and averages them to minimize noise variance.

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